

	Case Name: Policy Numbering System	Sector	IT
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule Risk Analysis	Schedule with resources Schedule with costs Random simulation One of nine std. scenarios User defined distributions
Submitted by	Matilde Casado and Margarida Antunes		
Date	2015	Project Control	Automatic tracking
File Name	C2015-20 Policy Numbering System		Tracking based on user input

1. Project description

Project authenticity

The Policy Numbering System project follows a structured development process from planning and analysis to design, development, implementation, and final deployment.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	11	Serial/Parallel (SP)	20%
Planned Duration (PD)	171 days*	Activity Distribution (AD)	62%
Budget At Completion (BAC)	12.645 €	Length of Arcs (LA)	20%
Renewable Resources	6	Topological Float (TF)	13%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity				Time sensitivity		
	avg [%]	std dev [%]	skew [-]		avg [%]	std dev [%]	skew [-]
CRI-r	16.8	26.7	-1.3	CI	27.4	35.7	-1.2
CRI-rho	38.9	24.0	1.0	SI	27.4	34.4	-0.9
CRI-tau	57.7	42.3	0.1	SSI	12.2	26.5	-2.6
				CRI-r	15.4	24.2	-1.9
				CRI-rho	37.8	22.1	0.2
				CRI-tau	59.0	41.0	0.0
	Resource sensitivity						
	avg [%]	std dev [%]	skew [-]				
CRI-r	13.1	22.4	-1.9				
CRI-rho	13.1	22.4	-1.9				
CRI-tau	13.1	22.4	-1.9				

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	2251.8	-2251.8	1	1.3	1.2
PV - SPI	2237.5	-2237.5	CPI	1.2	1.2
PV - SCI	2145.9	-2145.9	SPI	1.8	1.8
ED - 1	2502.9	-2502.9	SPI(t)	2.6	2.6
ED - SPI	2237.5	-2237.5	SCI	1.7	1.7
ED - SCI	2149.8	-2149.8	SCI(t)	2.5	2.5
ES - 1	2482.9	-2482.9	0.8 CPI + 0.2 SPI	1.3	1.3
ES - SPI(t)	1931.8	-1931.8	0.8 CPI + 0.2 SPI(t)	1.5	1.5
ES - SCI(t)	1877.0	-1877.0			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 4 tracking periods with a length of approximately 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	1586.0	-566.5	-423.3	1.2	1.0	0.3	1.0
std dev	39.8	651.5	415.6	0.0	0.1	0.4	0.0
final	1545.0	0.0	0.0	1.1	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	171 days	Real Duration	161 days	Early	0%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	190.3	0.5	4.5	-4.5
PV - SPI	190.8	1.0	4.6	-4.6
PV - SCI	191.5	1.3	4.7	-4.7
ED - 1	192.5	1.3	4.9	-4.9
ED - SPI	193.5	1.3	5.0	-5.0
ED - SCI	194.3	1.0	5.2	-5.2
ES - 1	194.8	0.5	5.2	-5.2
ES - SPI(t)	195.3	0.5	5.3	-5.3
ES - SCI(t)	195.8	1.0	5.4	-5.4

2.3.3.3. Cost forecasting

BAC	12.645 €	Real Cost	11.100 €	Under Budget	-12.2%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	11059.0	39.8	0.1	0.1
CPI	10979.2	133.2	0.3	0.3
SPI	11115.0	42.2	0.0	0.0
SPI(t)	14955.1	4566.0	-8.7	-8.7
SCI	11027.2	70.3	0.2	0.2
SCI(t)	14339.0	3882.7	-7.3	-7.3
0.8 CPI + 0.2 SPI	11001.3	106.0	0.2	0.2
0.8 CPI + 0.2 SPI(t)	11079.8	29.0	0.0	0.0